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1.Alert Classification

Objective

The objective of this task is to understand how security alerts are categorized and prioritized in a Security Operations Center (SOC). Alerts are analyzed based on severity, potential impact, and associated attack techniques.

Alert Classification Table

Alert ID	Alert Type	Priority	MITRE Technique
001	Phishing Email	High	T1566
002	Brute Force SSH	Medium	T1110
003	Malware Detected	Critical	T1204
004	Port Scan	Low	T1046

Explanation

Security alerts are classified based on their potential impact on the organization.
Critical alerts represent severe threats such as ransomware infections or system compromise. High-priority alerts indicate possible unauthorized access or phishing attempts. Medium alerts represent suspicious activity that requires investigation, while low-priority alerts include reconnaissance activities like port scanning.

1.1 Alert Prioritization using CVSS

SOC analysts use the Common Vulnerability Scoring System (CVSS) to determine the severity of vulnerabilities and prioritize alerts accordingly. CVSS scores range from 0.0 to 10.0, where higher scores indicate more critical vulnerabilities.

Alert	CVSS Score	Priority
Log4Shell Exploit	9.8	Critical
SQL Injection	8.5	High
Brute Force Login	6.5	Medium
Port Scan	3.1	Low

This prioritization helps SOC teams focus on the most critical threats first.

2. Alert Triage

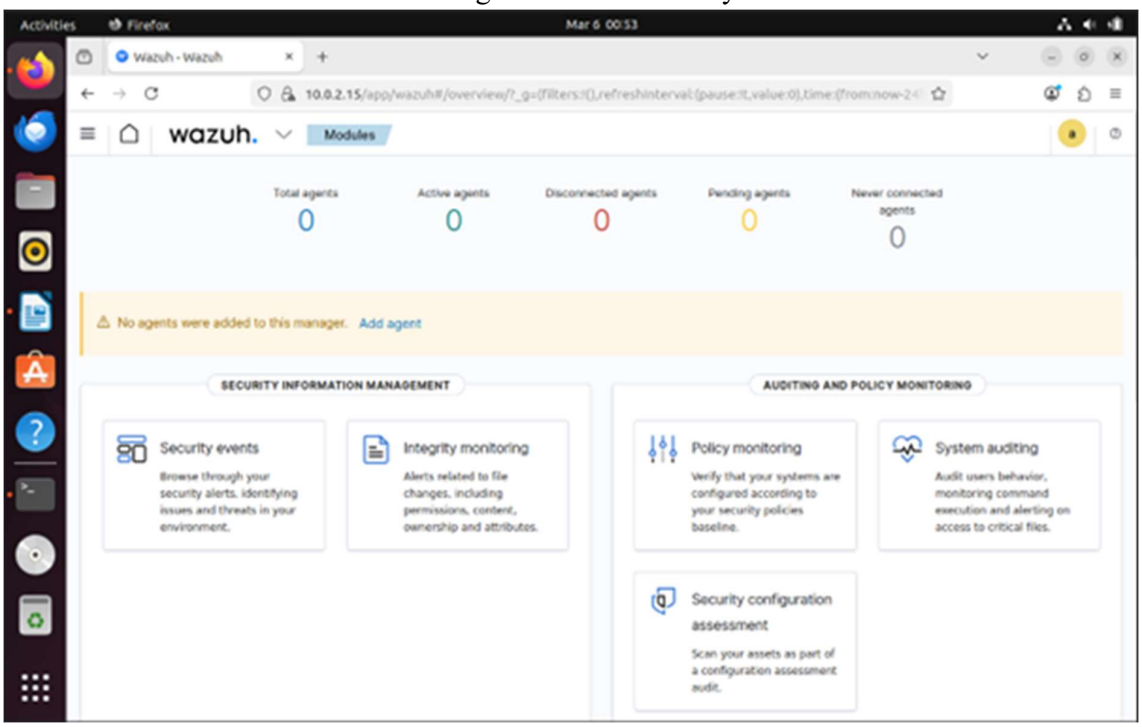
Objective

Alert triage is the process of analyzing security alerts to determine their severity and validity. SOC analysts review alerts and determine whether they represent a real threat or a false positive.

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH attempts	192.168.1.100	Medium	Open
003	Suspicious login activity	10.0.0.5	High	Investigating

Investigation Steps

1. Review security logs in the Wazuh dashboard.
2. Verify suspicious IP addresses using threat intelligence platforms such as VirusTotal.
3. Check indicators of compromise using AlienVault OTX.
4. Escalate confirmed threats to higher-level SOC analysts.



Threat Intelligence Validation

The suspicious IP address 192.168.1.100 was analyzed using the AlienVault OTX threat intelligence platform. The analysis revealed that this IP address has been associated with multiple brute-force login attempts targeting Linux servers.

Based on this threat intelligence information, the alert was considered potentially malicious and further monitoring was recommended.

3. Incident Response

Objective

Incident response involves identifying, containing, and resolving security incidents to minimize damage to the organization.

Executive Summary

A suspicious login activity was detected by the Wazuh security monitoring system. The event was flagged as a potential brute-force attack attempt targeting a system within the monitored environment.

Incident Timeline

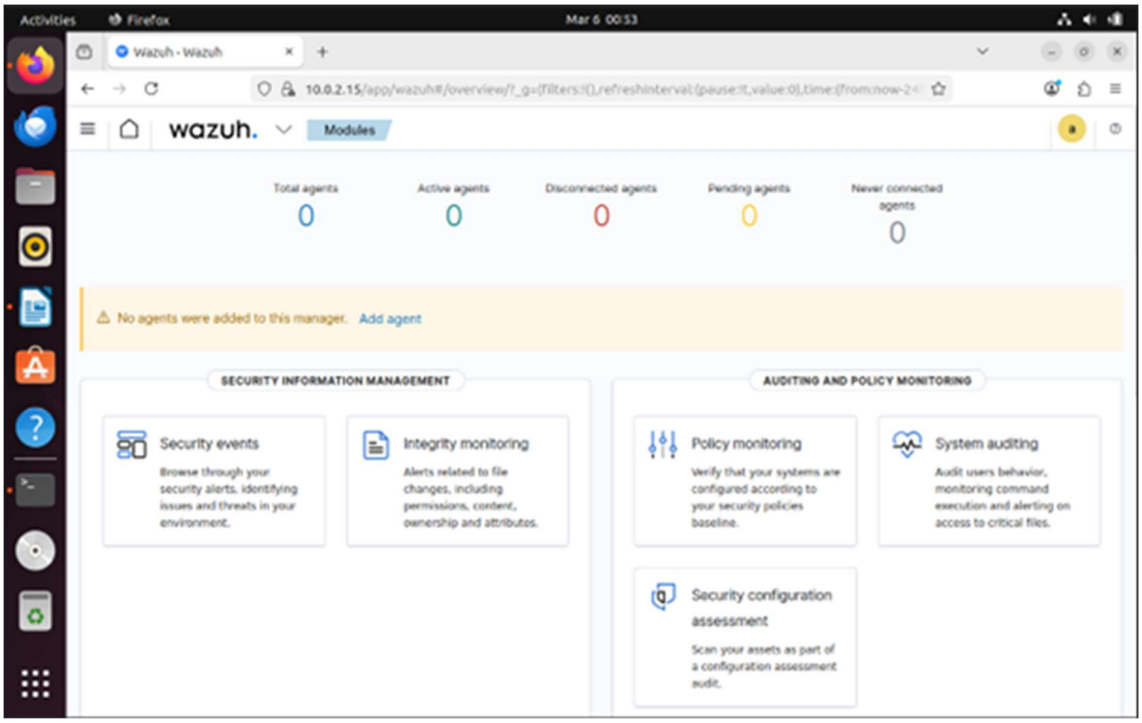
Time	Action
10:00 AM	Alert detected by Wazuh
10:05 AM	SOC analyst reviewed logs
10:10 AM	Suspicious IP investigated
10:15 AM	Endpoint isolated

Remediation Steps

- Block the malicious IP address using firewall rules.
- Reset compromised credentials.
- Update system security policies.
- Monitor the system for further suspicious activities.

Lessons Learned

Early detection through SIEM monitoring significantly improves the response time and reduces the impact of security incidents.



4. Incident Ticket

Incident Details

Title: ***[High Priority] Suspicious Login Attempt Detected***

Description:

Multiple login attempts were detected from an unknown IP address. The activity triggered an alert in the monitoring system indicating a possible brute-force attack.

Indicators of Compromise:

IP Address: 192.168.1.100

Event Type: Brute Force Login Attempt

Priority Level: High

Assigned To: SOC Analyst

Status: Under Investigation

4.1 Escalation Email

Subject: Security Alert Escalation – Suspicious Login Attempt

Dear Tier 2 Security Team,

A high-priority security alert has been detected involving multiple suspicious login attempts from IP address 192.168.1.100. The activity indicates a possible brute-force attack targeting the monitored system.

Initial investigation has been conducted and the affected endpoint has been isolated to prevent further unauthorized access. Additional analysis is required to determine the attack source and scope of compromise.

Please review the incident and assist with advanced investigation.

Regards
SOC Analyst

5. Evidence Preservation

Objective

Evidence preservation ensures that digital evidence is collected and stored securely without modification so it can be used during investigations.

Evidence Log

Item	Description	Collected By	Date	Hash Value
Alert Logs	Wazuh alerts.json file	SOC Analyst	2026-03-06	SHA256 Hash

Procedure

1. Identify relevant log files containing security alerts.
2. Copy logs from the monitored system.
3. Generate a cryptographic hash to ensure integrity.
4. Store evidence in a secure repository.

```
┌──(kali㉿kali)-[~/Desktop/wazuh-docker/single-node]
└─$ sudo docker exec -it single-node-wazuh.manager-1 bash
root@wazuh:/# logger "SOC alert test generated"
root@wazuh:/# exit
exit

┌──(kali㉿kali)-[~/Desktop/wazuh-docker/single-node]
└─$ sudo docker exec -it single-node-wazuh.manager-1 bash
root@wazuh:/# sha256sum /var/ossec/logs/alerts/alerts.json
9756aefbf8c86e0d684682b9472157c9df8bb5958194ccd7a1f3e2e8f41d1067  /var/ossec/logs/alerts/alerts.json
root@wazuh:/#
```

6. Wazuh Monitoring Dashboard

Tool Used

Security monitoring was performed using **Wazuh**, deployed using **Docker** on **Kali Linux**.

Purpose

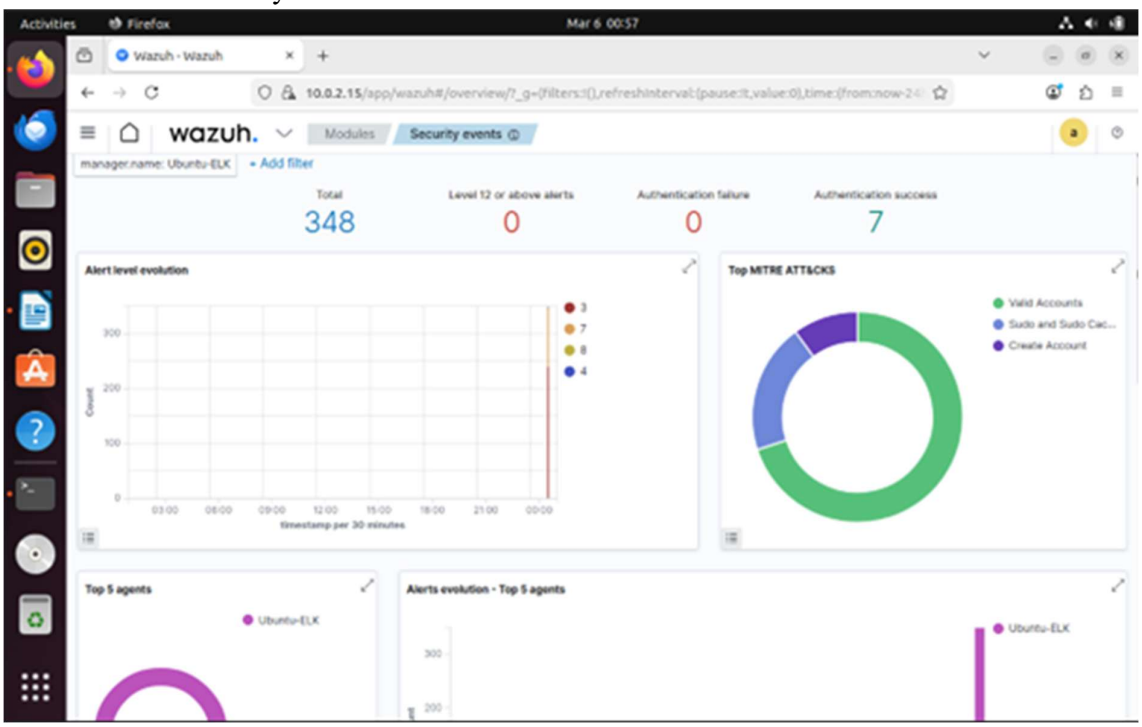
Wazuh provides real-time security monitoring by analyzing logs and detecting suspicious activity within a system.

Observations

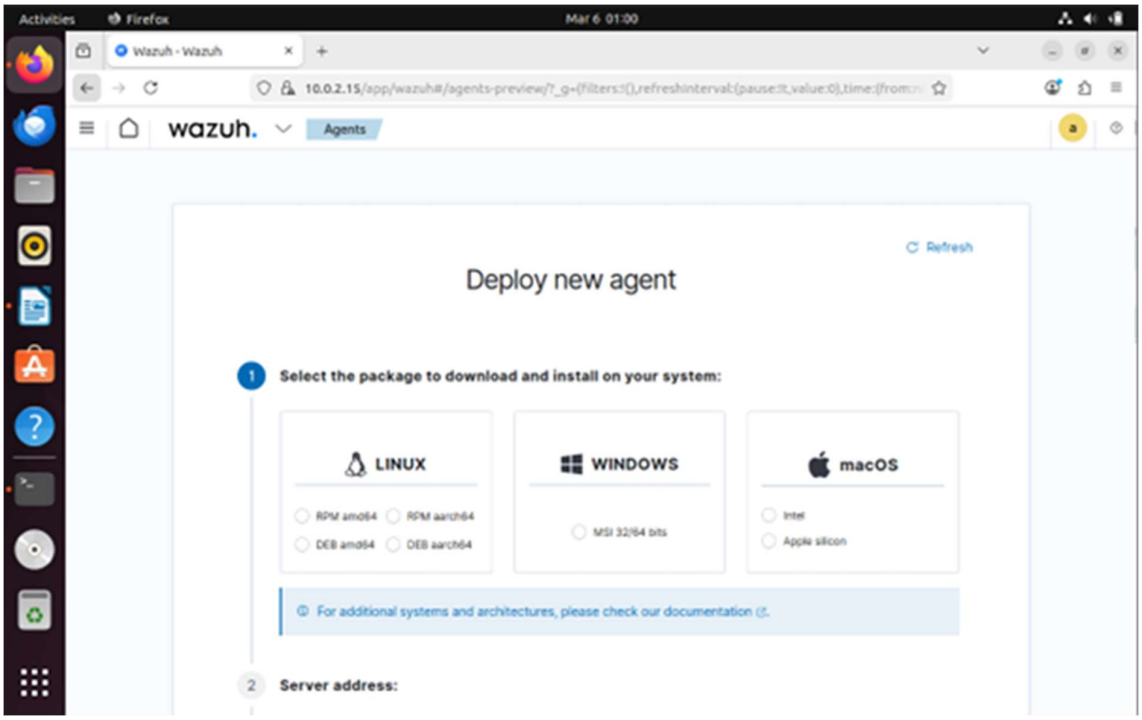
The dashboard displays security alerts, system monitoring information, and event logs. It helps SOC analysts quickly identify and respond to potential threats.

Wazuh Dashboard Overview

1. Wazuh Security Events



2. Wazuh Agents Page



7. Conclusion

This exercise demonstrated the core responsibilities of a SOC analyst, including alert monitoring, incident classification, investigation, and response. The Wazuh SIEM platform was used to monitor system activity and detect potential security threats. Proper alert triage and incident response procedures help organizations minimize security risks and improve overall cyber defense capabilities.